Self-Check

Lecture 0

Prerequisites: A Refresher

This sheet checks whether you are fluent in what the first lectures rely on: complex arithmetic, matrix algebra, determinants and Gaussian elimination. Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

First solve problems **0.1, 0.2, 0.9, 0.15, 0.22, 0.23** and **0.30** *without notes*. Then check your solutions against the answer sheet, which you will find on Moodle.

After that, work only where gaps showed: reread the corresponding section of Lecture 0 and solve the remaining problems of that section. If everything was correct, solve as many of the remaining problems as you like for practice.

Complex arithmetic

Exercise 0.1 (diagnostic) Let $z = 3 + 2i$ and $w = 1 + i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z + w = 4 + 3i$; $\quad zw = 1 + 5i$; $\quad z/w = \frac{5}{2} - \frac{1}{2}i$; $\quad \overline{z} = 3 - 2i$; $\quad |z| = \sqrt{13}$.

Exercise 0.2 (diagnostic) Write $1 - i$ in polar form and use it to compute $(1 - i)^8$.

Answer. $1 - i = \sqrt{2}\, e^{-i\pi/4}$, so $(1 - i)^8 = 16$.

Exercise 0.3 (computational) Find all complex numbers z with $z^2 = i$.

Hint. Write i in polar form and take square roots: halve the argument, and remember a quadratic has two roots.

Answer. $z = \pm\dfrac{1+i}{\sqrt{2}} = \pm\, e^{i\pi/4}$.

Exercise 0.4 (computational) Let $z = 3 + i$ and $w = 1 + 2i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z + w = 4 + 3i$; $\quad zw = 1 + 7i$; $\quad z/w = 1 - i$; $\quad \overline{z} = 3 - i$; $\quad |z| = \sqrt{10}$.

Exercise 0.5 (computational) Write $1 + i$ in polar form and use it to compute $(1 + i)^6$.

Answer. $1 + i = \sqrt{2}\, e^{i\pi/4}$, so $(1 + i)^6 = -8i$.

Exercise 0.6 (computational) Write $\sqrt{3} + i$ in polar form and use it to compute $(\sqrt{3} + i)^4$, giving the answer in Cartesian form.

Answer. $\sqrt{3} + i = 2\, e^{i\pi/6}$, so $(\sqrt{3} + i)^4 = -8 + 8\sqrt{3}\, i$.

Exercise 0.7 (bridge) For complex numbers $z = a + bi$ and $w = c + di$, prove that $\overline{zw} = \overline{z}\,\overline{w}$.

Hint. Multiply the two numbers first and take the conjugate of the product. Then take the conjugate of each factor and multiply them. Either way, two lines of real arithmetic suffice.

Exercise 0.8 (extension) Find all four complex numbers z with $z^4 = -4$.

Hint. Write -4 in polar form as $4\, e^{i\pi}$; a fourth root has modulus $4^{1/4} = \sqrt{2}$ and argument $\frac{1}{4}(\pi + 2\pi k)$ for $k = 0, 1, 2, 3$.

Answer. $z \in \{1 + i,\ -1 + i,\ -1 - i,\ 1 - i\}$, i.e. $z = \pm 1 \pm i$.

Matrix algebra

Exercise 0.9 (diagnostic) Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Compute AB, BA, and A^{T}, and verify $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ even though $AB \neq BA$.

Answer. $AB = \begin{pmatrix} 5 & 5 \\ -1 & -2 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix}$, $A^{\mathsf{T}} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$; $\mathrm{tr}(AB) = 3 = \mathrm{tr}(BA)$.

Exercise 0.10 (computational) Let

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

Compute AB and BA, and verify $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ although $AB \neq BA$.

Answer. $AB = \begin{pmatrix} 2 & -2 \\ 7 & 2 \end{pmatrix}$, $BA = \begin{pmatrix} 1 & -3 \\ 5 & 3 \end{pmatrix}$; $\mathrm{tr}(AB) = 4 = \mathrm{tr}(BA)$.

Exercise 0.11 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, compute A^2 and A^3.

Answer. $A^2 = \begin{pmatrix} 4 & 4 \\ 0 & 4 \end{pmatrix}$, $A^3 = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$.

Exercise 0.12 (bridge) Let $A = \begin{pmatrix} 1+i & 2-i \\ 0 & 3i \end{pmatrix}$ over $\mathbb{C}$. Write down the transpose A^{T} and the conjugate transpose $A^{\dagger} = \overline{A}^{\mathsf{T}}$.

Answer. $A^{\mathsf{T}} = \begin{pmatrix} 1+i & 0 \\ 2-i & 3i \end{pmatrix}$, $A^{\dagger} = \begin{pmatrix} 1-i & 0 \\ 2+i & -3i \end{pmatrix}$.

Exercise 0.13 (extension) Prove that the cyclic identity $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ holds for square matrices A, B of the same size.

Hint. Write both traces as double sums over entries and note that the middle step only reorders a finite sum of scalars.

Exercise 0.14 (bridge) For matrices $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times p}(\mathbb{F})$, prove that $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$.

Hint. Compare the (i, j) entries of the two sides. Write $(AB)_{ji}$ as a sum and notice that, after transposing, its factors appear in the opposite order.

Determinants

Exercise 0.15 (diagnostic) Evaluate $\det\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 1$; invertible.

Exercise 0.16 (computational) Use the 2×2 formula to find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$.

Exercise 0.17 (bridge) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find all λ for which $\det(\lambda I - A) = 0$. In Lecture 5 we will call these values the eigenvalues of A.

Answer. $\lambda = 1$ and $\lambda = 3$.

Exercise 0.18 (computational) Evaluate $\det \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 4$; invertible.

Exercise 0.19 (computational) Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$. Compute $\det A$, $\det B$, and $\det(AB)$, and check that $\det(AB) = \det A \ \det B$.

Answer. $\det A = -2$, $\det B = 6$, $\det(AB) = -12 = (-2)(6)$.

Exercise 0.20 (extension) Evaluate the block-diagonal determinant

$$\det \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}.$$

Hint. If one of the off-diagonal blocks is zero, the determinant equals the product of the determinants of the diagonal blocks.

Answer. $\det = 6$.

Exercise 0.21 (extension) Prove that $\det(AB) = \det A \ \det B$ for any 2×2 matrices A, B. Compute directly from the entries.

Hint. Multiply AB out and expand $\det(AB)$. Four of the terms cancel in pairs, and what is left factors as $(ad - bc)(eh - fg)$.

Gaussian elimination

Exercise 0.22 (diagnostic) Solve the system by Gaussian elimination:

$$\begin{aligned} x + \ y + \ z &= 2, \\ x - \ y + 2z &= 5, \\ 2x + \ y - \ z &= 2. \end{aligned}$$

Answer. $(x, y, z) = (2, -1, 1)$ (unique).

Exercise 0.23 (diagnostic) Find a basis and the dimension of the solution space of the homogeneous system $Ax = 0$, that is, of the null space of A, where $A = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 3 & 1 \end{pmatrix}$.

Hint. Row-reduce A, identify the pivot and free columns, then solve for the pivot variables in terms of the free ones.

Answer. Basis $\{(1,1,0,0),\ (-2,0,1,1)\}$; dimension 2.

Exercise 0.24 (computational) Find the inverse matrix A^{-1} by row-reducing the augmented matrix $[A \mid I]$ with Gaussian elimination, for $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

Exercise 0.25 (computational) Solve the system by Gaussian elimination:

$$\begin{aligned} x + \ \ y + \ \ z &= 6, \\ x - \ \ y + \ \ z &= 2, \\ 2x + \ \ y - \ \ z &= 1. \end{aligned}$$

Answer. $(x, y, z) = (1, 2, 3)$ (unique).

Exercise 0.26 (computational) Write the set of all solutions of the system

$$\begin{aligned} x + 2y - \ \ z &= 3, \\ 2x + \ \ y + \ \ z &= 3, \end{aligned}$$

in the form "particular solution + any vector in the null space".

Answer. $(x, y, z) = (1, 1, 0) + t(-1, 1, 1)$, $t \in \mathbb{R}$.

Exercise 0.27 (computational) Find the inverse matrix A^{-1} by row-reducing the augmented matrix $[A \mid I]$ with Gaussian elimination, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 0.28 (bridge) Let $A \in M_{m\times n}(\mathbb{F})$. Prove that the null space $\ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is closed under addition and scalar multiplication. In Lecture 1 this will mean that $\ker A$ is a subspace of $\mathbb{F}^n$.

Hint. Matrix multiplication is linear in x, so $A(x + y)$ and $A(cx)$ split at once. Check $0 \in \ker A$ first.

Exercise 0.29 (extension) For which real values of a does the system

$$\begin{aligned} ax + \ \ y + \ \ z &= 1, \\ x + ay + \ \ z &= 1, \\ x + \ \ y + az &= 1 \end{aligned}$$

have a unique solution, infinitely many solutions, or no solution? When the solution is unique, write it down.

Hint. Add the three equations to get $(a+2)(x+y+z) = 3$; subtract pairs to get $(a-1)(x-y) = 0$ and $(a-1)(y-z) = 0$. The determinant of the coefficient matrix is $(a-1)^2(a+2)$.

Answer. Unique iff $a \neq 1$ and $a \neq -2$, with $x = y = z = \dfrac{1}{a+2}$; infinitely many if $a = 1$ (the plane $x + y + z = 1$); no solution if $a = -2$.

Exercise 0.30 (diagnostic) Classify each displayed RREF augmented matrix as having no solution, a unique solution, or infinitely many solutions. For each consistent system, give its full solution set.

$$\text{(a)}\ \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right], \qquad \text{(b)}\ \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right], \qquad \text{(c)}\ \left[\begin{array}{ccc|c} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Answer. (a) Unique: $(2,-1)$. (b) No solution. (c) Infinitely many: $(x,y,z) = (4,-1,0) + t(2,-3,1)$, $t \in \mathbb{R}$.

Exercise 0.31 (repair) Row-reduce and classify each system, giving the full solution set when it is consistent:

$$\begin{array}{lll} \text{(a)} & x+y=1, & x-y=3, \\ \text{(b)} & x+y=1, & 2x+2y=3, \\ \text{(c)} & x+y=1, & 2x+2y=2. \end{array}$$

Answer. (a) Unique: $(2,-1)$. (b) No solution. (c) Infinitely many: $(x,y) = (1,0) + t(-1,1)$, $t \in \mathbb{R}$.

Analysis check

These five problems test not algebra but what you remember from your calculus course. Lecture 0 does not need them, but solve them before the lectures in which we apply linear algebra to differential equations, series and Fourier coefficients.

Exercise 0.32 (analysis route) Differentiate $f(t) = t^2 e^{-t}$.

Answer. $f'(t) = e^{-t}(2t - t^2)$.

Exercise 0.33 (analysis route) Use integration by parts to evaluate $\displaystyle\int_0^1 te^t\,dt$.

Answer. 1.

Exercise 0.34 (analysis route) Find the general real solution of $y'' - 3y' + 2y = 0$.

Answer. $y(t) = C_1 e^t + C_2 e^{2t}$.

Exercise 0.35 (analysis route) Determine whether the series $\displaystyle\sum_{n=0}^{\infty} 3^{-n}$ converges and, if it does, find its sum.

Answer. It converges to $\dfrac{3}{2}$.

Exercise 0.36 (analysis route) For $f(x) = x$ on $[-\pi, \pi]$, compute the first Fourier sine coefficient $b_1 = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} x \sin x\,dx$.

Answer. $b_1 = 2$.